



Properties

$$E(0) = B(0) = H(0) = S(0) = C(0) = 1$$

$$K = \begin{bmatrix} \begin{array}{c} \text{MildSteelTemperate} \\ \text{MildSteelColdWeather} \\ \text{AH36} \\ \text{DH36} \\ \text{EH36} \\ \text{EH40} \\ \text{Al5083} \\ \text{Al5086} \\ \text{GRP} \\ \text{CFRP} \\ \text{TiGrade5} \end{array} & \begin{array}{cc} T^+ & T^- \end{array} & \begin{array}{c} k_D \\ k_H \\ k_\sigma \\ k_U \end{array} \end{bmatrix}$$

$$K = \begin{bmatrix} & T^+ & T^- & D & H & \sigma & U \\ E & k_{E,T^+} & k_{E,T^-} & k_{E,D} & 0 & 0 & 0 \\ B & k_{B,T^+} & k_{B,T^-} & 0 & 0 & 0 & k_{B,U} \\ H & k_{H,T^+} & k_{H,T^-} & 0 & 0 & 0 & k_{H,U} \\ S & 0 & 0 & k_{S,D} & k_{S,H} & 0 & k_{S,U} \\ C & 0 & 0 & 0 & k_{C,H} & k_{C,\sigma} & 0 \end{bmatrix}$$

Environmental Factors

1. Extreme heat

1. Overview
 1. Engine: oil thins, cooling system overloads, intake air is less dense reducing power
 2. Battery: permanent capacity reduction
 3. Hydraulics: fluid thins, seals degrade
2. Input
 1. $T(t)$ - temperature as function of time
3. Output
 1. $\frac{dE}{dt} = -k_E \cdot \max(T(t) - T_E, 0)$
 2. $\frac{dB}{dt} = -k_B \cdot \max(T(t) - T_B, 0)$
 3. $\frac{dH}{dt} = -k_H \cdot \max(T(t) - T_H, 0)$

2. Extreme cold

1. Overview
 1. Engine: cold starts fail, oil thickens, fuel gels
 2. Battery: immediate capacity drop, charge acceptance drops
 3. Hydraulics: fluid sluggish, actuators fail
2. Input
 1. $T(t)$ - temperature as function of time
3. Output
 1. $\frac{dE}{dt} = -k_E \cdot \max(T_E - T(t), 0)$
 2. $\frac{dB}{dt} = -k_B \cdot \max(T_B - T(t), 0)$

$$3. \quad \frac{dH}{dt} = -k_H \cdot \max(T_H - T(t), 0)$$

3. Dust ingestion

1. Overview
 1. Engine: intake filters clog, abrasive particles score cylinder walls
 2. Sensors: optical surfaces scratched and obscured, radar antenna elements clogged
2. Input
 1. $D(t)$ - particulate concentration as function of time
3. Output
 1. $\frac{dE}{dt} = -k_E \cdot D(t)$
 2. $\frac{dS}{dt} = -k_S \cdot D(t)$

4. Humidity

1. Overview
 1. Sensors: condensation on optics, corrosion on electrical contacts
 2. Chassis: rust at weld points and joints, seal degradation on hatches
2. Input
 1. $Y(t)$ - relative humidity as function of time
3. Output
 1. $\frac{dS}{dt} = -k_S \cdot \max(Y(t) - Y_S, 0)$
 2. $\frac{dC}{dt} = -k_C \cdot \max(Y(t) - Y_C, 0)$

5. Salinity

1. Overview
 1. Chassis: accelerated corrosion at welds, joints, underbelly
2. Input
 1. $\sigma(t)$ - salinity concentration as function of time
3. Output
 1. $\frac{dC}{dt} = -k_C \cdot \sigma(t)^2$

6. UV / Solar radiation

1. Overview

1. Sensors: optical coating degradation, accuracy drift
 2. Hydraulics: external hoses become brittle
 3. Battery: permanent capacity reduction from sustained heat load
2. Input
 1. $U(t)$ - irradiance as function of time
 3. Output
 1. $\frac{dS}{dt} = -k_S \cdot U(t)$
 2. $\frac{dH}{dt} = -k_H \cdot U(t)$
 3. $\frac{dB}{dt} = -k_B \cdot U(t)^2$

Vehicle Health

$$V(t) = \frac{4}{13} \cdot E(t) + \frac{3}{13} \cdot B(t) + \frac{3}{13} \cdot H(t) + \frac{1}{13} \cdot S(t) + \frac{2}{13} \cdot C(t)$$